

Logistic regression

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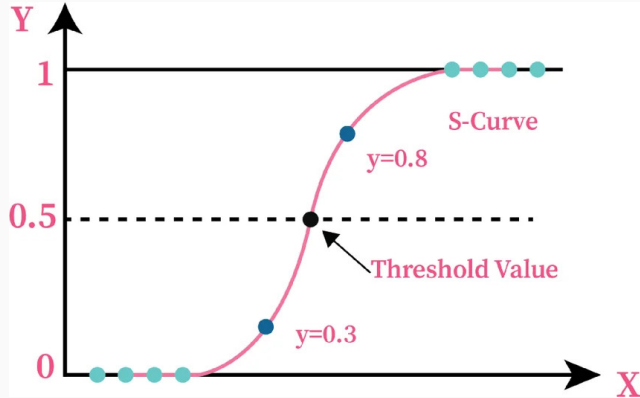
- Linear regression involves:
 - Working with the numerical outcome
 - Find the best fitted line/plane to describe the relationship between the IVs and the DV
- Logistic regression involves:
 - Working with the categorical outcome
 - Find the best fitted logistic curve (sigmoid curve) to describe the relationship between the IVs and the probability of the outcome/DV

- There are several types of logistic regression:
 - Binary logistic regression: the categorical outcome is 2 levels (our focus)
 - Polytomous /multinomial logistic regression: the categorical outcome is > 2 levels
 - Ordinal logistic regression: the categorical outcome is > 2 levels and has a clear natural order (example: mild, moderate, severe)

Simple logistic regression

- Aims: to describe the a relationship between 1 binary categorical DV and 1 numerical/categorical IV
- When to conduct:
 - Random sample (theoretical assumption)
 - Binary outcome
- Examples:
 - What is the effect of age on getting sick (yes/no)?
 - What is the relationship between price and whether the item is getting bought in a month or not?

- The main idea of logistic regression is to model the probability of a binary outcome based on the IV - sigmoid curve



- Assumptions:

1. Linearity: each continuous IV with the logit transformed DV (use Box-Tidwell method)

**Box-Tidwell method (example: $age \times \ln(age)$):*

1. *Natural log transformed the continuous IVs*
2. *Create interaction term between the natural log transformed of the variable with its original variable*

- Model fit:

1. Hosmer-lemeshow test - fit if $p > 0.05$

- Check the model's predicted probability match the reality (compare the distribution of the predicted probabilities and the observed values)

2. Classification table - fit if $> 60\%$

- Force the model to make a prediction and calculate how many times the model is right

3. Area under the receiver operating characteristic (ROC) curve - fit if $> 60\%$

- Reflects the ability of the model to separate the groups (the outcome level)

Simple logistic regression: Analysis

- Background: we want investigate the impact of GPA on the admission to a graduate school (admitted/not admitted)
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Check the model fit
 4. Interpretation
 5. Conclusion

- Step 1:
 - H_0 : There is no relationship between GPA and a graduate school admission ($\beta_1 = 0$)
 - H_A : There is a relationship between GPA and a graduate school admission ($\beta_1 \neq 0$)

Packages

```
library(tidyverse) #data manipulation  
library(readstata13) #read stata data  
library(QuantPsyc) #classification table  
library(ResourceSelection) #hosmer-lemeshow test  
library(pROC) #roc-auc
```

Load in the data

```
log_data <- read.dta13("Data/logistic_regression.dta")
```

First 6 rows of the data

```
head(log_data) %>% knitr::kable()
```

admit	gre	gpa	rank
0	380	3.61	3
1	660	3.67	3
1	800	4.00	1
1	640	3.19	4
0	520	2.93	4
1	760	3.00	2

Run simple logistic regression analysis

```
slogr_mod <- glm(admit ~ gpa, family = binomial(), data = log_data)
```

- Step 2: Assumptions:

1. Linearity of cont IV with logit transformed DV

```
log_data <-
```

```
  log_data %>%
```

```
  mutate(LNgpa = log(gpa) * gpa)
```

```
update(slogr_mod, .~. + LNgpa) %>%
```

```
  summary()
```

- P value > 0.05 indicate a linearity is assumed

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.288948	14.897278	0.2207751	0.8252676
gpa	-4.004291	9.839179	-0.4069741	0.6840270
LNgpa	2.281941	4.442076	0.5137104	0.6074545

- Step 3: Model fit:
 - Classification table - > 0.6 , so, model is fit

```
ctable <- ClassLog(slogr_mod, slogr_mod$y)
ctable$overall
```

```
[1] 0.6825
```

- Step 3: Model fit:
 - Hosmer-lemeshow test - p value > 0.05 , so, model is fit

```
hoslem.test(slogr_mod$y, fitted(slogr_mod), g=10)
```

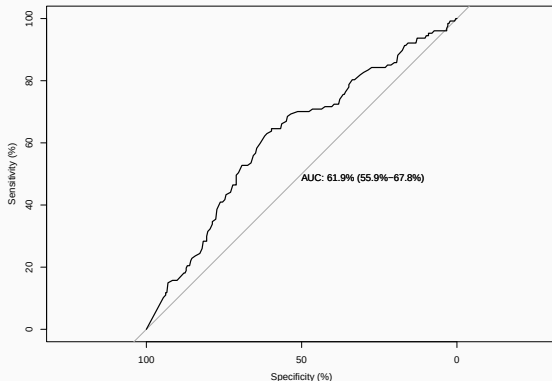
Hosmer and Lemeshow goodness of fit (GOF) test

```
data: slogr_mod$y, fitted(slogr_mod)
```

```
X-squared = 9.3281, df = 8, p-value = 0.3154
```

- Step 3: Model fit:
 - ROC-AUC - $> 60\%$, so, model is fit

```
roc_curve <- roc(slogr_mod$y, fitted(slogr_mod), ci = T, percent = T)
plot(roc_curve, print.auc = T)
```



- Step 4: Interpret:
 - P value < 0.05
 - 95% CI does not cross 0
 - There is a significant relationship between GPA and a graduate school admission

P value

```
summary(slogr_mod)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.357588	1.0353170	-4.208940	2.565712e-05
gpa	1.051109	0.2988694	3.516951	4.365350e-04

OR and 95% CI

```
cbind(exp(slogr_mod$coefficients), exp(confint(slogr_mod))) %>%  
  round(digits = 2)
```

		2.5 %	97.5 %
(Intercept)	0.01	0.00	0.09
gpa	2.86	1.61	5.20

- Step 4: Interpret:
 - Those with higher GPA had 2.86 higher odds of being admitted to a graduate school
-

Odds ratio (OR):

- *Ratio of odds of an event in one group vs. the odds of the event in the other group*
- *Measure the association between an exposure and an outcome*
- *Odds*
 - *Likelihood of an event to occur*
 - *Odds = $P / (1-P)$, P is probability of an event to occur*
- *Interpretation:*
 - *$OR=1$ IV does not affect odds of outcome*
 - *$OR>1$ IV associated with higher odds of outcome*
 - *$OR<1$ IV associated with lower odds of outcome*

- Step 5: Conclusion:
 - There is a relationship between GPA and a graduate school admission

Multiple logistic regression

- Aims: to describe a relationship between a binary categorical outcome variable and several independent variables
- When to conduct:
 - Random sample (theoretical assumption)
 - Binary outcome
- Examples:
 - What are the factors influencing the diagnosis of diabetes?
 - What are the factors influencing whether a political candidate winning an election?
 - What are the factors influencing the admission to a graduate school? (our data)

- Assumptions (similar to simple logistic regression):
 1. Linearity: each continuous IV with the logit transformed DV (use Box-Tidwell method)

Multiple logistic regression: Analysis

- Background: we want investigate the factors influencing the admission to a graduate school
- Steps (we have extra steps similar to multiple linear regression):
 1. State the hypothesis
 2. Check for the interaction
 3. Check the assumptions
 4. Check the outliers and influential observations
 5. Check the multicollinearity
 6. Check the model fit
 7. Interpretation
 8. Conclusion

- Step 1:
 - H_0 : There is no relationship between GPA, GRE result, rank, and a graduate school admission ($\beta_1 = 0$)
 - H_A : There is a relationship between GPA, GRE result, rank, and a graduate school admission ($\beta_1 \neq 0$)

Packages

```
library(tidyverse) #data manipulation  
library(readstata13) #read stata data  
library(QuantPsyc) #classification table  
library(ResourceSelection) #hosmer-lemeshow test  
library(pROC) #roc-auc
```

Load in the data

```
log_data <- read.dta13("Data/logistic_regression.dta") %>%  
  mutate(  
    # Add a new variable (dummy)
```

First 6 rows of the data

```
head(log_data) %>% knitr::kable()
```

admit	gre	gpa	rank
0	380	3.61	3
1	660	3.67	3
1	800	4.00	1
1	640	3.19	4
0	520	2.93	4
1	760	3.00	2

Run multiple logistic regression analysis

```
mlogr <- glm(admit ~ gre + gpa + rank, family = binomial(),  
             data = log_data)
```

- Step 2: Interaction: all p value > 0.05 (not significant)

```
int_mod1 <- update(mlogr, .~. + gre * gpa)
summary(int_mod1)$coefficients[7, , drop = FALSE]
```

	Estimate	Std. Error	z value	Pr(> z)
gre:gpa	-0.004718986	0.002897741	-1.628505	0.1034178

```
int_mod2 <- update(mlogr, .~. + gre * rank)
summary(int_mod2)$coefficients[7:9, , drop = FALSE]
```

	Estimate	Std. Error	z value	Pr(> z)
gre:rank2	-0.0012411874	0.002835410	-0.43774535	0.6615709
gre:rank3	-0.0001126133	0.003024849	-0.03722941	0.9703021
gre:rank4	-0.0004107883	0.003632166	-0.11309734	0.9099534

```
int_mod3 <- update(mlogr, .~. + gpa * rank)
summary(int_mod3)$coefficients[7:9, , drop = FALSE]
```

	Estimate	Std. Error	z value	Pr(> z)
gpa:rank2	-0.342950656	0.8584403	-0.399504387	0.6895216
gpa:rank3	-0.515244580	0.9292811	-0.554455051	0.5792675
gpa:rank4	-0.009245855	1.2120947	-0.007627997	0.9939138

- Step 3: Assumptions:

1. Linearity of cont IV with logit transformed DV

```
log_data <-  
  log_data %>%  
    mutate(LNgpa = log(gpa) * gpa,  
           LNgre = log(gre) * gre)  
  
update(mlogr, .~. + LNgpa + LNgre) %>%  
  summary()
```

- P value > 0.05 indicate a linearity is assumed

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.080404537	15.926413865	0.005048502	0.9959718950
gre	0.024707138	0.062820017	0.393300410	0.6940976156
gpa	-3.037533768	10.269978334	-0.295768274	0.7674070530
rank2	-0.679449370	0.317402394	-2.140656096	0.0323017815
rank3	-1.342417873	0.345629985	-3.883973987	0.0001027629
rank4	-1.545034778	0.418283622	-3.693749162	0.0002209718
LNgpa	1.732384979	4.639484186	0.373400341	0.7088505055
LNgre	-0.003044543	0.008513111	-0.357629893	0.7206203043

- Step 4: Outliers and influential observations

```
out_data <- cooks.distance(mlogr)
sum(out_data > 1)
```

```
[1] 0
```

- Step 5: Multicollinearity

```
car::vif(mlogr) #all GVIF^(1/(2*Df)) should be < 3.16
```

	GVIF	Df	GVIF^(1/(2*Df))
gre	1.134377	1	1.065071
gpa	1.155902	1	1.075129
rank	1.025759	3	1.004248

- Step 6: Model fit:
 - Classification table - > 0.6 , so, model is fit

```
ctable <- ClassLog(mlogr, mlogr$y)
ctable$overall
```

```
[1] 0.71
```

- Step 6: Model fit:
 - Hosmer-lemeshow test - p value > 0.05 , so, model is fit

```
hoslem.test(mlogr$y, fitted(mlogr), g=10)
```

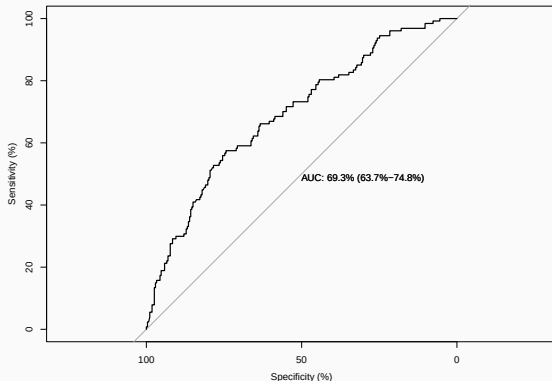
Hosmer and Lemeshow goodness of fit (GOF) test

```
data: mlogr$y, fitted(mlogr)
```

```
X-squared = 11.085, df = 8, p-value = 0.1969
```

- Step 6: Model fit:
 - ROC-AUC - $> 60\%$, so, model is fit

```
roc_curve <- roc(mlogr$y, fitted(mlogr), ci = T, percent = T)  
plot(roc_curve, print.auc = T)
```



- Step 7: Interpret:
 - P value < 0.05
 - 95% CI does not cross 0
 - There is a significant relationship between all included factors and a graduate school admission

P value

```
summary(mlogr)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.989979375	1.139950942	-3.500133	0.0004650269
gre	0.002264426	0.001093998	2.069863	0.0384651364
gpa	0.804037654	0.331819297	2.423119	0.0153878837
rank2	-0.675442930	0.316489662	-2.134171	0.0328288186
rank3	-1.340203930	0.345306419	-3.881202	0.0001039415
rank4	-1.551463677	0.417831634	-3.713131	0.0002047107

OR and 95% CI

```
cbind(exp(mlogr$coefficients), exp(confint(mlogr))) %>%  
  round(digits = 3)
```

		2.5 %	97.5 %
(Intercept)	0.019	0.002	0.167
gre	1.002	1.000	1.004
gpa	2.235	1.174	4.324
rank2	0.509	0.272	0.945
rank3	0.262	0.132	0.512
rank4	0.212	0.091	0.471

- Step 7: Interpret:
 - Those with higher GRE result had 0.2% higher odds of being admitted to a graduate school after adjusting for other variables
 - Those with higher GPA result had 2.24 higher odds of being admitted to a graduate school after adjusting for other variables
 - Those in rank 2 50% lower odds of being admitted to a graduate school compared to those in rank 1 after adjusting for other variables
 - Those in rank 3 73.8% lower odds of being admitted to a graduate school compared to those in rank 1 after adjusting for other variables
 - Those in rank 4 78.8% lower odds of being admitted to a graduate school compared to those in rank 1 after adjusting for other variables

- Step 8: Conclusion:
 - There is a significant relationship between all included factors and a graduate school admission